

Young-mok Jung, Ph.D

Machine Learning Research Engineer

 bit.ly/youngmokjung
 San Diego, CA

WORK EXPERIENCE

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| OCT 2025 – PRESENT | Apple Inc.
<i>Machine Learning Engineer</i>
San Diego, CA |
| | <ul style="list-style-type: none">• LLM in Agentic System: Developing and evaluating planner module in agentic systems (LLM-Siri) to handle complex, multi-step user queries. Also implemented harness-agnostic memory evaluation framework focusing on personal assistant task domain.• RL-post training for Vision QA: Implementing Reinforcement Learning (RL) pipelines to optimize reward models and evaluation rubrics for multi-modal Vision Question Answering tasks. |
| AUG 2023 – SEP 2025 | Inocras Inc. - bioinformatics startup for precision oncology
<i>AI & Data Team Lead / AI Tech Lead</i>
San Diego, CA |
| | <ul style="list-style-type: none">• Genomic Foundation Model: Spearheaded the development of <i>DNACHUNKER</i> (arXiv:2601.03019), a learnable tokenization strategy for DNA language models. Led collaborative research to train and evaluate foundation models on proprietary multi-modal datasets for cancer subtyping.• MRD Vision: Led design, implementation, and clinical validation of an ML-based Minimal Residual Disease (MRD) detection pipeline, used to evaluate therapeutic efficacy in drug development. Achieved ppm-level sensitivity in collaboration with Ultima Genomics.• HPC & Cloud Optimization: Re-architected the cancer genome analysis workflow for cloud efficiency by eliminating centralized disk dependencies, enabling vertical/horizontal scalability, integrating hardware acceleration, and optimizing CPU, memory, and storage utilization. Reduced runtime by 50% and costs by 80%, driving a 10%+ product margin increase.• CNS Tumor Classification: Designed an ML pipeline for whole-genome methylation-based classification of CNS tumors, translating biological domain knowledge into computational objectives. |
| SEP 2018 – FEB 2024 | KAIST
<i>Graduate Research Assistant (Advisor: Prof. Dongsu Han & Prof. Young Seok Ju)</i>
South Korea |
| | <ul style="list-style-type: none">• Transfer Learning for Deep Variant Callers: Developed a domain adaptation framework for DeepVariant and Clair3. Improved SNP/INDEL F1-scores by 6.4%p / 9.4%p and maintained accuracy with 50% less labeled data (Preprint '23).• BWA-MEME (Bioinformatics Acceleration): Designed memory and CPU-cache efficient alignment software using learned indexes. Reduced CPU instructions by 4.6x and LLC misses by 2.2x, achieving 3.45x speedup over Intel BWA-MEM2 (Bioinformatics '22, cited 270 times).• LiveNAS: Built a live video streaming system using online training for super-resolution DNNs. Reduced bandwidth usage by 45.9% while maintaining quality (SIGCOMM '20, cited 172 times). |

EDUCATION

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| SEP 2018 – FEB 2024 | KAIST
Ph.D. in Electrical Engineering (AI/ML Systems for Bioinformatics)
Dissertation: <i>Enhancing genome analysis pipeline with AI and ML</i>
Daejeon, Korea |
| FEB 2014 – AUG 2018 | KAIST
B.S. in Electrical Engineering
Daejeon, Korea |

SKILLS

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| GENERATIVE AI & RL | Foundation Models, Agentic system, Transformers, RL, PyTorch |
| ML SYSTEMS & HPC | CPU&Memory profiling, SIMD/C++, RDMA/TCP
Implemented algorithms to optimize custom transport protocols that reduce latency in datacenter or accelerate alignment software in bioinformatics. |
| COMPUTATIONAL BIO | Alignment, Mutation Calling, Minimal Residual Disease detection, Methylation sequencing
Designed clinical-grade pipelines for cancer detection and drug efficacy assessment. |

PUBLICATIONS

[P1]	DNACHUNKER: Learnable Tokenization for DNA Language Models <i>T. Kim, J. Shin, H. Kim, Youngmok Jung, et al.</i>	(to appear) ICML '26
[A1]	Improving minimal residual disease detection: A tumor-informed whole-genome approach Youngmok Jung , S. Ferguson, S. Lee, et al.	ASCO '25
[P2]	Enabling whole-genome DNA methylation-based classification of central nervous system tumors <i>J. Lee, Y. Ju, B. Oh, ... Corresponding: Youngmok Jung</i>	medRxiv '25
[C1]	TopFull: An Adaptive Top-Down Overload Control for SLO-Oriented Microservices <i>J. Park, J. Park, Youngmok Jung, H. Lim, H. Yeo, D. Han</i>	SIGCOMM '24
[P3]	Generalizing deep variant callers via domain adaptation and semi-supervised learning Youngmok Jung , J. Park, H. Lim, J. Lee, Y. Ju, D. Han	Preprint '23
[J1]	BWA-MEME: BWA-MEM emulated with a machine learning approach Youngmok Jung , D. Han	Bioinformatics '22
[C2]	Co-optimizing for Flow Completion Time in Radio Access Network <i>J. Kim, Y. Lee, H. Lim, Youngmok Jung, S. Kim, D. Han</i>	CoNEXT '22
[C3]	Engorgio: Neural Video Enhancement at Scale <i>H. Yeo, H. Lim, J. Kim, Youngmok Jung, J. Ye, D. Han</i>	SIGCOMM '22
[C4]	Towards Timeout-less Transport in Commodity Datacenter Networks <i>H. Lim, W. Bai, Y. Zhu, Youngmok Jung, D. Han</i>	EuroSys '21
[C5]	Neural-Enhanced Live Streaming: Improving Live Video Ingest via Online Learning <i>J. Kim, Youngmok Jung (Co-First), H. Yeo, J. Ye, D. Han</i>	SIGCOMM '20
[C6]	Enabling Neural-enhanced Video Streaming on Commodity Mobile Devices <i>H. Yeo, C. Chong, Youngmok Jung, J. Ye, D. Han</i>	MobiCom '20
[C7]	Neural Adaptive Content-aware Internet Video Delivery <i>H. Yeo, Youngmok Jung, J. Kim, J. Shin, D. Han</i>	OSDI '18

AWARDS & HONORS

Feb 2023	Samsung 29th Humantech Paper Award (Silver Prize) Communication & Networks Track
Feb 2022	Samsung PhD Sponsorship Selected for full research funding and tuition support
Sep 2020	1st Place, Kiwoom US Stock Trading Competition Achieved 201% ROI (Ranked 1st out of 10,000+ participants)

OPEN SOURCE & PROJECTS

MAIN AUTHOR	BWA-MEME (ML-Accelerated Aligner)  ★ 136 🍴 34k+ Pulls Optimized C++ genome aligner using ML based index structures. Achieved 3.45x speedup over Intel BWA-MEM2. Widely used in production pipelines via Bioconda/Docker.
MAIN AUTHOR	RUN-DVC (Deep Variant Caller)  PyTorch Semi-supervised domain adaptation framework for genomic variant callers (DeepVariant, Clair3). Enables accurate mutation calling on new sequencing platforms with limited data.
CONTRIBUTOR	Collaborative Systems Projects  C++, NS-3, TensorFlow Contributed to high-performance networking and video systems: NeuroScaler (Server-side video enhancement), NEMO (Mobile super-resolution), and TLT-RDMA (Datacenter transport).